

BleedSense — Supplementary Materials

Computer Vision and Cognitive Systems (CVCS) course, A.Y. 2025/2026, UNIMORE — Riccardo Mazzi (294437). Full tables for every experimental phase and every extension, all architectures and conditions (the main report only reports summary averages). Mean $\pm$ standard deviation over the 5 HemoSet folds, single run for Rabbani (fixed split).

Table 1 — Phase 1: In-Domain vs Cross-Dataset Baseline

Setting	Architecture	DICE	IOU	PRECISION	RECALL	F1	HD95
HemoSet $\rightarrow$ HemoSet (in-domain)	unet	0.738 $\pm$ 0.035	0.607 $\pm$ 0.038	0.779 $\pm$ 0.056	0.726 $\pm$ 0.037	0.738 $\pm$ 0.035	25.852 $\pm$ 5.863
HemoSet $\rightarrow$ Rabbani (cross)	unet	0.218 $\pm$ 0.033	0.149 $\pm$ 0.025	0.638 $\pm$ 0.081	0.179 $\pm$ 0.036	0.218 $\pm$ 0.033	134.346 $\pm$ 20.055
Rabbani $\rightarrow$ Rabbani (in-domain)	unet	0.689	0.561	0.745	0.711	0.689	49.819
Rabbani $\rightarrow$ HemoSet (cross)	unet	0.588	0.442	0.505	0.752	0.588	62.789
HemoSet $\rightarrow$ HemoSet (in-domain)	unetplusplus	0.749 $\pm$ 0.026	0.617 $\pm$ 0.031	0.732 $\pm$ 0.034	0.791 $\pm$ 0.046	0.749 $\pm$ 0.026	32.914 $\pm$ 18.234
HemoSet $\rightarrow$ Rabbani (cross)	unetplusplus	0.236 $\pm$ 0.035	0.159 $\pm$ 0.023	0.418 $\pm$ 0.096	0.223 $\pm$ 0.022	0.236 $\pm$ 0.035	104.057 $\pm$ 8.325
Rabbani $\rightarrow$ Rabbani (in-domain)	unetplusplus	0.675	0.550	0.731	0.712	0.675	54.827
Rabbani $\rightarrow$ HemoSet (cross)	unetplusplus	0.597	0.455	0.529	0.728	0.597	56.133
HemoSet $\rightarrow$ HemoSet (in-domain)	deeplabv3plus	0.748 $\pm$ 0.027	0.617 $\pm$ 0.026	0.775 $\pm$ 0.050	0.751 $\pm$ 0.055	0.748 $\pm$ 0.027	27.977 $\pm$ 11.522
HemoSet $\rightarrow$ Rabbani (cross)	deeplabv3plus	0.196 $\pm$ 0.032	0.133 $\pm$ 0.025	0.665 $\pm$ 0.074	0.165 $\pm$ 0.035	0.196 $\pm$ 0.032	151.249 $\pm$ 20.629
Rabbani $\rightarrow$ Rabbani (in-domain)	deeplabv3plus	0.689	0.558	0.740	0.709	0.689	39.340
Rabbani $\rightarrow$ HemoSet (cross)	deeplabv3plus	0.569	0.420	0.476	0.762	0.569	57.188

Table 2 — Phase 2: Data Augmentation (light vs aggressive)

Setting	Architecture	Augmentation	DICE	IOU	PRECISION	RECALL	F1	HD95
HemoSet → HemoSet (in-domain)	unet	light	0.738±0.035	0.607±0.038	0.779±0.056	0.726±0.037	0.738±0.035	25.852±5.863
HemoSet → Rabbani (cross)	unet	light	0.218±0.033	0.149±0.025	0.638±0.081	0.179±0.036	0.218±0.033	134.346±20.055
Rabbani → Rabbani (in-domain)	unet	light	0.689	0.561	0.745	0.711	0.689	49.819
Rabbani → HemoSet (cross)	unet	light	0.588	0.442	0.505	0.752	0.588	62.789
HemoSet → HemoSet (in-domain)	unet	aggressive	0.756±0.033	0.629±0.035	0.752±0.049	0.794±0.025	0.756±0.033	26.835±5.965
HemoSet → Rabbani (cross)	unet	aggressive	0.323±0.032	0.227±0.027	0.396±0.071	0.370±0.031	0.323±0.032	90.396±4.508
Rabbani → Rabbani (in-domain)	unet	aggressive	0.672	0.546	0.732	0.699	0.672	38.526
Rabbani → HemoSet (cross)	unet	aggressive	0.603	0.461	0.539	0.747	0.603	51.224
HemoSet → HemoSet (in-domain)	unetplusplus	light	0.749±0.026	0.617±0.031	0.732±0.034	0.791±0.046	0.749±0.026	32.914±18.234
HemoSet → Rabbani (cross)	unetplusplus	light	0.236±0.035	0.159±0.023	0.418±0.096	0.223±0.022	0.236±0.035	104.057±8.325
Rabbani → Rabbani (in-domain)	unetplusplus	light	0.675	0.550	0.731	0.712	0.675	54.827
Rabbani → HemoSet (cross)	unetplusplus	light	0.597	0.455	0.529	0.728	0.597	56.133
HemoSet → HemoSet (in-domain)	unetplusplus	aggressive	0.756±0.022	0.629±0.025	0.793±0.053	0.753±0.020	0.756±0.022	28.457±8.364
HemoSet → Rabbani (cross)	unetplusplus	aggressive	0.304±0.028	0.212±0.022	0.349±0.019	0.367±0.051	0.304±0.028	97.351±7.258
Rabbani → Rabbani (in-domain)	unetplusplus	aggressive	0.701	0.580	0.734	0.741	0.701	43.259
Rabbani → HemoSet (cross)	unetplusplus	aggressive	0.615	0.475	0.546	0.749	0.615	60.491
HemoSet → HemoSet (in-domain)	deeplabv3plus	light	0.748±0.027	0.617±0.026	0.775±0.050	0.751±0.055	0.748±0.027	27.977±11.522
HemoSet → Rabbani (cross)	deeplabv3plus	light	0.196±0.032	0.133±0.025	0.665±0.074	0.165±0.035	0.196±0.032	151.249±20.629
Rabbani → Rabbani (in-domain)	deeplabv3plus	light	0.689	0.558	0.740	0.709	0.689	39.340
Rabbani → HemoSet (cross)	deeplabv3plus	light	0.569	0.420	0.476	0.762	0.569	57.188
HemoSet → HemoSet (in-domain)	deeplabv3plus	aggressive	0.730±0.039	0.599±0.045	0.736±0.061	0.757±0.030	0.730±0.039	29.543±11.536

Setting	Architecture	Augmentation	DICE	IOU	PRECISION	RECALL	F1	HD95
HemoSet → Rabbani (cross)	deeplabv3plus	aggressive	0.337±0.021	0.239±0.019	0.401±0.047	0.436±0.045	0.337±0.021	93.988±7.533
Rabbani → Rabbani (in-domain)	deeplabv3plus	aggressive	0.671	0.539	0.689	0.747	0.671	42.393
Rabbani → HemoSet (cross)	deeplabv3plus	aggressive	0.585	0.438	0.471	0.835	0.585	51.325

Table 3 — Phase 3: Appearance/Domain Adaptation (Reinhard, FDA)

All runs in this phase use "light" augmentation (to isolate the effect of adaptation alone); the `aggressive/none` rows are shown for comparison with Phase 2.

Setting	Architecture	Augmentation	Adaptation	DICE	IOU	PRECISION	RECALL	F1	HD95
HemoSet → HemoSet (in-domain)	unet	light	none	0.738±0.035	0.607±0.038	0.779±0.056	0.726±0.037	0.738±0.035	25.852±5.863
HemoSet → Rabbani (cross)	unet	light	none	0.218±0.033	0.149±0.025	0.638±0.081	0.179±0.036	0.218±0.033	134.346±20.055
Rabbani → Rabbani (in-domain)	unet	light	none	0.689	0.561	0.745	0.711	0.689	49.819
Rabbani → HemoSet (cross)	unet	light	none	0.588	0.442	0.505	0.752	0.588	62.789
HemoSet → HemoSet (in-domain)	unet	light	reinhard	0.709±0.081	0.573±0.096	0.735±0.157	0.727±0.060	0.709±0.081	31.177±20.222
HemoSet → Rabbani (cross)	unet	light	reinhard	0.259±0.044	0.176±0.032	0.397±0.111	0.299±0.110	0.259±0.044	97.635±14.636
Rabbani → Rabbani (in-domain)	unet	light	reinhard	0.640	0.519	0.758	0.649	0.640	62.700
Rabbani → HemoSet (cross)	unet	light	reinhard	0.589	0.453	0.541	0.718	0.589	57.643
HemoSet → HemoSet (in-domain)	unet	light	fda	0.709±0.074	0.575±0.087	0.773±0.052	0.683±0.104	0.709±0.074	33.596±18.038
HemoSet → Rabbani (cross)	unet	light	fda	0.241±0.038	0.165±0.028	0.474±0.079	0.221±0.069	0.241±0.038	108.179±15.021
Rabbani → Rabbani (in-domain)	unet	light	fda	0.686	0.563	0.763	0.694	0.686	53.186
Rabbani → HemoSet (cross)	unet	light	fda	0.578	0.445	0.533	0.680	0.578	62.265
HemoSet → HemoSet (in-domain)	unet	aggressive	none	0.756±0.033	0.629±0.035	0.752±0.049	0.794±0.025	0.756±0.033	26.835±5.965
HemoSet → Rabbani (cross)	unet	aggressive	none	0.323±0.032	0.227±0.027	0.396±0.071	0.370±0.031	0.323±0.032	90.396±4.508
Rabbani → Rabbani (in-domain)	unet	aggressive	none	0.672	0.546	0.732	0.699	0.672	38.526
Rabbani → HemoSet (cross)	unet	aggressive	none	0.603	0.461	0.539	0.747	0.603	51.224
HemoSet → HemoSet (in-domain)	unetplusplus	light	none	0.749±0.026	0.617±0.031	0.732±0.034	0.791±0.046	0.749±0.026	32.914±18.234
HemoSet → Rabbani (cross)	unetplusplus	light	none	0.236±0.035	0.159±0.023	0.418±0.096	0.223±0.022	0.236±0.035	104.057±8.325
Rabbani → Rabbani (in-domain)	unetplusplus	light	none	0.675	0.550	0.731	0.712	0.675	54.827
Rabbani → HemoSet (cross)	unetplusplus	light	none	0.597	0.455	0.529	0.728	0.597	56.133

Setting	Architecture	Augmentation	Adaptation	DICE	IOU	PRECISION	RECALL	F1	HD95
HemoSet → HemoSet (in-domain)	unetplusplus	light	reinhard	0.735±0.036	0.602±0.042	0.760±0.050	0.739±0.040	0.735±0.036	31.047±12.715
HemoSet → Rabbani (cross)	unetplusplus	light	reinhard	0.283±0.040	0.190±0.027	0.365±0.055	0.338±0.094	0.283±0.040	99.263±11.659
Rabbani → Rabbani (in-domain)	unetplusplus	light	reinhard	0.621	0.501	0.759	0.622	0.621	63.865
Rabbani → HemoSet (cross)	unetplusplus	light	reinhard	0.580	0.440	0.521	0.711	0.580	54.977
HemoSet → HemoSet (in-domain)	unetplusplus	light	fda	0.724±0.044	0.591±0.050	0.759±0.070	0.728±0.095	0.724±0.044	27.940±9.329
HemoSet → Rabbani (cross)	unetplusplus	light	fda	0.252±0.026	0.171±0.020	0.374±0.102	0.277±0.064	0.252±0.026	99.942±10.204
Rabbani → Rabbani (in-domain)	unetplusplus	light	fda	0.672	0.539	0.664	0.757	0.672	48.767
Rabbani → HemoSet (cross)	unetplusplus	light	fda	0.513	0.367	0.392	0.802	0.513	74.231
HemoSet → HemoSet (in-domain)	unetplusplus	aggressive	none	0.756±0.022	0.629±0.025	0.793±0.053	0.753±0.020	0.756±0.022	28.457±8.364
HemoSet → Rabbani (cross)	unetplusplus	aggressive	none	0.304±0.028	0.212±0.022	0.349±0.019	0.367±0.051	0.304±0.028	97.351±7.258
Rabbani → Rabbani (in-domain)	unetplusplus	aggressive	none	0.701	0.580	0.734	0.741	0.701	43.259
Rabbani → HemoSet (cross)	unetplusplus	aggressive	none	0.615	0.475	0.546	0.749	0.615	60.491
HemoSet → HemoSet (in-domain)	deeplabv3plus	light	none	0.748±0.027	0.617±0.026	0.775±0.050	0.751±0.055	0.748±0.027	27.977±11.522
HemoSet → Rabbani (cross)	deeplabv3plus	light	none	0.196±0.032	0.133±0.025	0.665±0.074	0.165±0.035	0.196±0.032	151.249±20.629
Rabbani → Rabbani (in-domain)	deeplabv3plus	light	none	0.689	0.558	0.740	0.709	0.689	39.340
Rabbani → HemoSet (cross)	deeplabv3plus	light	none	0.569	0.420	0.476	0.762	0.569	57.188
HemoSet → HemoSet (in-domain)	deeplabv3plus	light	reinhard	0.713±0.051	0.575±0.061	0.730±0.105	0.731±0.060	0.713±0.051	34.398±14.164
HemoSet → Rabbani (cross)	deeplabv3plus	light	reinhard	0.247±0.034	0.169±0.026	0.394±0.106	0.262±0.072	0.247±0.034	106.050±7.321
Rabbani → Rabbani (in-domain)	deeplabv3plus	light	reinhard	0.571	0.452	0.774	0.568	0.571	83.954
Rabbani → HemoSet (cross)	deeplabv3plus	light	reinhard	0.561	0.416	0.469	0.763	0.561	51.960
HemoSet → HemoSet (in-domain)	deeplabv3plus	light	fda	0.728±0.029	0.594±0.028	0.760±0.050	0.723±0.027	0.728±0.029	30.026±10.805
HemoSet → Rabbani (cross)	deeplabv3plus	light	fda	0.234±0.032	0.160±0.025	0.526±0.068	0.204±0.045	0.234±0.032	115.712±13.686

Setting	Architecture	Augmentation	Adaptation	DICE	IOU	PRECISION	RECALL	F1	HD95
Rabbani → Rabbani (in-domain)	deeplabv3plus	light	fda	0.666	0.537	0.692	0.704	0.666	41.395
Rabbani → HemoSet (cross)	deeplabv3plus	light	fda	0.560	0.421	0.480	0.713	0.560	61.675
HemoSet → HemoSet (in-domain)	deeplabv3plus	aggressive	none	0.730±0.039	0.599±0.045	0.736±0.061	0.757±0.030	0.730±0.039	29.543±11.536
HemoSet → Rabbani (cross)	deeplabv3plus	aggressive	none	0.337±0.021	0.239±0.019	0.401±0.047	0.436±0.045	0.337±0.021	93.988±7.533
Rabbani → Rabbani (in-domain)	deeplabv3plus	aggressive	none	0.671	0.539	0.689	0.747	0.671	42.393
Rabbani → HemoSet (cross)	deeplabv3plus	aggressive	none	0.585	0.438	0.471	0.835	0.585	51.325

Table 4a — Phase 4: Natural Joint Training vs Single-Source

Test set	Architecture	Training	DICE	IOU	PRECISION	RECALL	F1	HD95
HemoSet test	unet	single-source (Phase 1)	0.738±0.035	0.607±0.038	0.779±0.056	0.726±0.037	0.738±0.035	25.852±5.863
HemoSet test	unet	natural joint (Phase 4)	0.747±0.041	0.620±0.044	0.756±0.045	0.769±0.051	0.747±0.041	25.614±8.944
Rabbani test	unet	single-source (Phase 1)	0.689	0.561	0.745	0.711	0.689	49.819
Rabbani test	unet	natural joint (Phase 4)	0.638±0.053	0.511±0.055	0.688±0.081	0.717±0.052	0.638±0.053	59.293±9.693
HemoSet test	unetplusplus	single-source (Phase 1)	0.749±0.026	0.617±0.031	0.732±0.034	0.791±0.046	0.749±0.026	32.914±18.234
HemoSet test	unetplusplus	natural joint (Phase 4)	0.757±0.036	0.630±0.039	0.729±0.051	0.814±0.038	0.757±0.036	30.395±8.579
Rabbani test	unetplusplus	single-source (Phase 1)	0.675	0.550	0.731	0.712	0.675	54.827
Rabbani test	unetplusplus	natural joint (Phase 4)	0.631±0.056	0.504±0.059	0.683±0.088	0.701±0.038	0.631±0.056	60.076±10.358
HemoSet test	deeplabv3plus	single-source (Phase 1)	0.748±0.027	0.617±0.026	0.775±0.050	0.751±0.055	0.748±0.027	27.977±11.522
HemoSet test	deeplabv3plus	natural joint (Phase 4)	0.721±0.053	0.590±0.055	0.759±0.077	0.734±0.075	0.721±0.053	29.856±9.249
Rabbani test	deeplabv3plus	single-source (Phase 1)	0.689	0.558	0.740	0.709	0.689	39.340
Rabbani test	deeplabv3plus	natural joint (Phase 4)	0.640±0.034	0.511±0.031	0.724±0.022	0.668±0.039	0.640±0.034	56.070±7.804

Table 4b — Phase 4b: Balanced Joint Training vs Single-Source

Test set	Architecture	Training	DICE	IOU	PRECISION	RECALL	F1	HD95
HemoSet test	unet	single-source (Phase 1)	0.738±0.035	0.607±0.038	0.779±0.056	0.726±0.037	0.738±0.035	25.852±5.863
HemoSet test	unet	balanced joint (Phase 4b)	0.742±0.047	0.613±0.048	0.753±0.071	0.766±0.064	0.742±0.047	29.085±9.659
Rabbani test	unet	single-source (Phase 1)	0.689	0.561	0.745	0.711	0.689	49.819
Rabbani test	unet	balanced joint (Phase 4b)	0.658±0.008	0.532±0.012	0.734±0.075	0.700±0.061	0.658±0.008	58.447±4.808
HemoSet test	unetplusplus	single-source (Phase 1)	0.749±0.026	0.617±0.031	0.732±0.034	0.791±0.046	0.749±0.026	32.914±18.234
HemoSet test	unetplusplus	balanced joint (Phase 4b)	0.743±0.036	0.612±0.038	0.721±0.064	0.795±0.050	0.743±0.036	35.014±11.039
Rabbani test	unetplusplus	single-source (Phase 1)	0.675	0.550	0.731	0.712	0.675	54.827
Rabbani test	unetplusplus	balanced joint (Phase 4b)	0.658±0.027	0.532±0.030	0.736±0.044	0.692±0.017	0.658±0.027	55.882±10.242
HemoSet test	deeplabv3plus	single-source (Phase 1)	0.748±0.027	0.617±0.026	0.775±0.050	0.751±0.055	0.748±0.027	27.977±11.522
HemoSet test	deeplabv3plus	balanced joint (Phase 4b)	0.731±0.048	0.599±0.047	0.746±0.069	0.748±0.051	0.731±0.048	29.748±10.083
Rabbani test	deeplabv3plus	single-source (Phase 1)	0.689	0.558	0.740	0.709	0.689	39.340
Rabbani test	deeplabv3plus	balanced joint (Phase 4b)	0.653±0.017	0.525±0.019	0.722±0.044	0.687±0.033	0.653±0.017	51.383±4.134

Table 5 — Extension: Ensemble of the 3 Architectures (on aggressive augmentation)

No additional training: averaged sigmoid probabilities of the 3 checkpoints already trained (UNet, UNet++, DeepLabV3+, aggressive condition).

Setting	DICE	IOU	PRECISION	RECALL	F1	HD95
HemoSet → HemoSet (in-domain, ensemble)	0.770±0.025	0.647±0.028	0.798±0.054	0.773±0.015	0.770±0.025	24.637±5.623
HemoSet → Rabbani (cross, ensemble)	0.343±0.014	0.245±0.012	0.433±0.030	0.379±0.029	0.343±0.014	91.457±4.161
Rabbani → Rabbani (in-domain, ensemble)	0.706	0.583	0.757	0.739	0.706	39.571
Rabbani → HemoSet (cross, ensemble)	0.623	0.481	0.544	0.777	0.623	49.629

Table 6 — Extension: "Blood-Index" Input Channel (on aggressive augmentation)

Extra R/(R+G+B) channel concatenated to the input, tested on top of the aggressive condition. Direct comparison with Table 2 (the "aggressive" column without the extra channel).

Setting	Architecture	Aggressive	Aggressive + blood-index
HemoSet → HemoSet (in-domain)	unet	0.756±0.033	0.741±0.039
HemoSet → Rabbani (cross)	unet	0.323±0.032	0.330±0.030
Rabbani → Rabbani (in-domain)	unet	0.672	0.693
Rabbani → HemoSet (cross)	unet	0.603	0.529
HemoSet → HemoSet (in-domain)	unetplusplus	0.756±0.022	0.745±0.042
HemoSet → Rabbani (cross)	unetplusplus	0.304±0.028	0.316±0.028
Rabbani → Rabbani (in-domain)	unetplusplus	0.701	0.701
Rabbani → HemoSet (cross)	unetplusplus	0.615	0.509
HemoSet → HemoSet (in-domain)	deeplabv3plus	0.730±0.039	0.746±0.041
HemoSet → Rabbani (cross)	deeplabv3plus	0.337±0.021	0.347±0.032
Rabbani → Rabbani (in-domain)	deeplabv3plus	0.671	0.660

Setting	Architecture	Aggressive	Aggressive + blood-index
Rabbani → HemoSet (cross)	deeplabv3plus	0.585	0.539

Slightly improves HemoSet→Rabbani but markedly worsens Rabbani→HemoSet across all 3 architectures — an unfavorable trade-off, negative outcome.

Table 7 — Extension: MixStyle (on light augmentation, no adaptation)

Style mixing on the encoder's feature maps (layer1/layer2), no target-domain images required. Direct comparison with Table 1 (Phase 1 baseline, same light/none condition).

Setting	Architecture	Without MixStyle	With MixStyle
HemoSet → HemoSet (in-domain)	unet	0.738±0.035	0.740±0.053
HemoSet → Rabbani (cross)	unet	0.218±0.033	0.198±0.045
Rabbani → Rabbani (in-domain)	unet	0.689	0.674
Rabbani → HemoSet (cross)	unet	0.588	0.577
HemoSet → HemoSet (in-domain)	unetplusplus	0.749±0.026	0.754±0.026
HemoSet → Rabbani (cross)	unetplusplus	0.236±0.035	0.218±0.031
Rabbani → Rabbani (in-domain)	unetplusplus	0.675	0.655
Rabbani → HemoSet (cross)	unetplusplus	0.597	0.548
HemoSet → HemoSet (in-domain)	deeplabv3plus	0.748±0.027	0.742±0.031
HemoSet → Rabbani (cross)	deeplabv3plus	0.196±0.032	0.194±0.024
Rabbani → Rabbani (in-domain)	deeplabv3plus	0.689	0.683
Rabbani → HemoSet (cross)	deeplabv3plus	0.569	0.559

Diffuse but small degradation on Rabbani in-domain and both cross-dataset directions, across all 3 architectures — no gain in any condition, negative outcome.

Table 8 — Extension: Test-Time BatchNorm Recalibration (AdaBN, on aggressive augmentation)

No additional training: BatchNorm statistics of the already-trained aggressive checkpoints are recalibrated with forward passes on unlabeled target-domain images.

Checkpoint	Direction	Dice without recalibration	Dice with recalibration
unet fold0	HemoSet→Rabbani	0.3392	0.3080
unet fold1	HemoSet→Rabbani	0.3312	0.3045
unet fold2	HemoSet→Rabbani	0.3465	0.2973
unet fold3	HemoSet→Rabbani	0.2661	0.2596
unet fold4	HemoSet→Rabbani	0.3309	0.3012
unetplusplus fold0	HemoSet→Rabbani	0.2719	0.2976
unetplusplus fold1	HemoSet→Rabbani	0.3068	0.3195
unetplusplus fold2	HemoSet→Rabbani	0.3419	0.3638
unetplusplus fold3	HemoSet→Rabbani	0.3154	0.3221
unetplusplus fold4	HemoSet→Rabbani	0.2816	0.3260
deeplabv3plus fold0	HemoSet→Rabbani	0.3379	0.3358
deeplabv3plus fold1	HemoSet→Rabbani	0.3387	0.3415
deeplabv3plus fold2	HemoSet→Rabbani	0.3039	0.3201
deeplabv3plus fold3	HemoSet→Rabbani	0.3621	0.3262
deeplabv3plus fold4	HemoSet→Rabbani	0.3447	0.3440
unet	Rabbani→HemoSet	0.6030	0.5708
unetplusplus	Rabbani→HemoSet	0.6145	0.5666

Checkpoint	Direction	Dice without recalibration	Dice with recalibration
deeplabv3plus	Rabbani→HemoSet	0.5845	0.5712

Direction	Phase	DICE	IOU	PRECISION	RECALL	F1	HD95
HemoSet → Rabbani (cross)	without recalibration	0.321±0.029	0.226±0.024	0.382±0.053	0.391±0.052	0.321±0.029	93.912±6.761
HemoSet → Rabbani (cross)	with test-time BN recalibration	0.318±0.025	0.218±0.020	0.283±0.033	0.546±0.036	0.318±0.025	104.253±6.685
Rabbani → HemoSet (cross)	without recalibration	0.601±0.015	0.458±0.019	0.519±0.041	0.777±0.051	0.601±0.015	54.347±5.321
Rabbani → HemoSet (cross)	with test-time BN recalibration	0.570±0.003	0.433±0.004	0.641±0.033	0.574±0.027	0.570±0.003	61.863±4.515

Shifts the precision/recall trade-off in opposite directions in the two cross-dataset directions, but net Dice does not improve in either direction —
neutral/negative outcome.

Final Table — Comparative Summary (averaged over 3 architectures)

Strategy	Phase	HemoSet Dice	HemoSet HD95	Rabbani Dice	Rabbani HD95	H→R Dice	H→R HD95	R→H Dice	R→H HD95
Baseline (source-only, light augmentation)	1	0.745	28.9	0.684	48.0	0.217	129.9	0.585	58.7
Aggressive augmentation	2	0.747	28.3	0.681	41.4	0.321	93.9	0.601	54.3
Appearance adaptation (Reinhard)	3	0.719	32.2	0.611	70.2	0.263	101.0	0.577	54.9
Appearance adaptation (FDA)	3	0.720	30.5	0.675	47.8	0.242	107.9	0.550	66.1
Natural joint training	4	0.742	28.6	0.636	58.5	n/a*	n/a*	n/a*	n/a*
Balanced joint training	4b	0.739	31.3	0.656	55.2	n/a*	n/a*	n/a*	n/a*
Blood-index channel (on aggressive)	extra	0.744	—†	0.685	—†	0.331	—†	0.526	—†
MixStyle (on light/none)	extra	0.745	—†	0.671	—†	0.203	—†	0.561	—†
Test-time BN recalibration (on aggressive)	extra	n/a**	n/a**	n/a**	n/a**	0.318	104.3	0.570	61.9
Ensemble of 3 architectures (on aggressive)	extra	0.770	24.6	0.706	39.6	0.343	91.5	0.623	49.6

* joint training is evaluated in-domain on both test sets (having seen both domains during training), so there is no true "cross-dataset" condition to report for this row. ** test-time BN recalibration adapts the statistics towards a specific target domain, so it does not produce a single static model that is also valid in-domain: for this reason it has no comparable in-domain value. † HD95 was not tabulated for this extension (see Table 6/7 above for the Dice-only comparison it was evaluated with).

Recommended configuration: aggressive augmentation in training + ensemble of the 3 architectures at inference — the only combination that improves cross-dataset robustness in both directions, with no appreciable in-domain cost. The three additional extensions (blood-index, MixStyle, test-time BN) do not beat it, but their analysis is discussed in the main report (Section 5.2).